FOOD COMPOSITION AND ADDITIVES

Modification to AOAC Official Methods 2009.01 and 2011.25 to Allow for Minor Overestimation of Low Molecular Weight Soluble Dietary Fiber in Samples Containing Starch

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AOAC Official Methods 2009.01 and 2011.25 have been modified to allow removal of resistant maltodextrins produced on hydrolysis of various starches by the combination of pancreatic α-amylase and amyloglucosidase (AMG) used in these assay procedures. The major resistant maltodextrin, 6^3,6^5-di-α-D-glucosyl maltopentaose, is highly resistant to hydrolysis by microbial α-glucosidases, isoamylase, pullulanase, pancreatic, bacterial and fungal α-amylase and AMG. However, this oligosaccharide is hydrolyzed by the mucosal α-glucosidase complex of the pig small intestine (which is similar to the human small intestine), and thus must be removed in the analytical procedure. Hydrolysis of these oligosaccharides has been by incubation with a high concentration of a purified AMG at 60°C. This incubation results in no hydrolysis or loss of other resistant oligosaccharides such as FOS, GOS, XOS, resistant maltodextrins (e.g., Fibersol 2) or polydextrose. The effect of this additional incubation with AMG on the measured level of low molecular weight soluble dietary fiber (SDFS) and of total dietary fiber in a broad range of samples is reported. Results from this study demonstrate that the proposed modification can be used with confidence in the measurement of dietary fiber.

Resistant starch and nondigestible oligosaccharides behave physiologically as dietary fiber, i.e., they are not digested or absorbed in the human small intestine but are utilized by the microbiota in the large intestine. Methods have been developed for the measurement of total dietary fiber (TDF) including high molecular weight dietary fiber (HMWDF) and low molecular weight dietary fiber (LMWDF; 1); for the separate measurement of insoluble dietary fiber (IDF), high molecular weight soluble dietary fiber (HMWSDF), and SDFS (1); and validated in interlaboratory studies under the auspices of AOAC INTERNATIONAL and AACC International as AOAC Official Methods **2009.01** (2) and **2011.25** (3), respectively. In recent nomenclature changes, HMWDF is defined as being composed of IDF and SDFP (soluble dietary fiber that precipitates in the presence of 76% aqueous ethanol) and SDFS (soluble dietary fiber that remains soluble in the presence of 76% aqueous ethanol; 2, 3). In AOAC Methods **2009.01** and **2011.25**, starch hydrolysis conditions were designed to obtain an accurate measure of resistant starch, i.e., the starch that is resistant to digestion in the human small intestine. Another aim was to accurately measure oligosaccharides that are resistant to digestion under physiological conditions. Samples are incubated with pancreatic α-amylase (PAA) and amyloglucosidase (AMG) under near physiological conditions (37°C, pH 6). This is followed by incubation with protease to hydrolyze the bulk of protein to peptides, allowing removal in the alcohol precipitation step to make gravimetric analysis of the residue more accurate and reproducible.

In these procedures it is essential to demonstrate that the enzymes used are sufficiently pure to give no hydrolysis and loss of nondigestible oligosaccharides such as fructo-oligosaccharides (FOS), galacto-oligosaccharides (GOS), xylosaccharides (XOS), resistant maltodextrins (RMD, e.g., Fibersol 2) or polydextrose.

Recent work by Brunt and Sanders (4) has shown that in particular samples high in starch, such as bread and partially cooked rice, a small percentage of soluble starch oligosaccharides resist hydrolysis by the PAA/AMG under the incubation conditions of AOAC Method **2009.01**. They found that on incubation of the SDFS fraction with a high concentration of AMG at elevated temperatures, some hydrolysis of the SDFS fraction occurred, with release of glucose.

These authors concluded that AOAC Methods **2009.01** and **2011.25** should be modified to include a further incubation of the SDFS fraction with AMG before HPLC. We repeated this work (5), and the results of Brunt and Sanders were confirmed. However, the level of AMG required to hydrolyze these RMDs is approximately 100-fold higher than that required to hydrolyze linear maltodextrins under similar incubation conditions. To study this further, the nonhydrolyzed maltodextrins were chromatographically purified and characterized by GC/MS and NMR (5) and were shown to be the heptasaccharide 6^3,6^5-di-α-D-glucosyl maltopentaose, with lesser amounts of the tetrasaccharide, 6^3-α-D-glucosyl maltotriose (Figure 1). The question that had to addressed was whether these oligosaccharides might actually be resistant to hydrolysis and absorption in the human small intestine. To answer this question, the mucosal α-glucosidase complex of a pig small intestine (which is similar to the human small intestine) was extracted according to the method of Dahlqvist (6), and the action on a range of malto-oligosaccharides (maltose, isomaltose, panose, 6^3-α-D-glucosyl maltotriose and 6^3,6^5-di-α-D-glucosyl

Received December 12, 2013.
The method was approved by the Expert Review Panel for Food Composition and Additives as First Action.
The Expert Review Panel for Food Composition and Additives invites method users to provide feedback on the First Action methods. Feedback from method users will help verify that the methods are fit for purpose and are critical to gaining global recognition and acceptance of the methods. Comments can be sent directly to the corresponding author or methodfeedback@aoac.org.
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DOI: 10.5740/jaoacint.13-406

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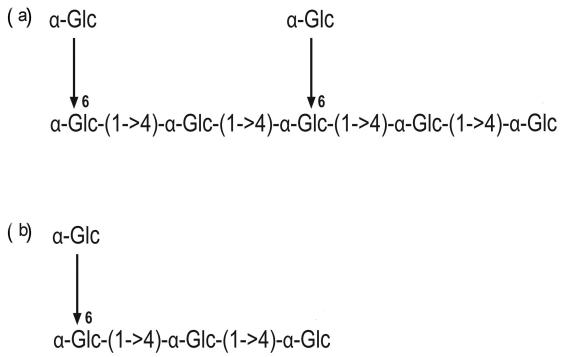

Figure 1. Structures of the two major branched maltodextrins that are resistant to hydrolysis under the incubation conditions of AOAC Method 2009.01. (a) 6^3,6^5-di-α-D-glucosyl maltopentaose; (b) 6^3-α-D-glucosyl maltotriose.

maltopentaose) was studied. In fact, the rates of hydrolysis of isomaltose, panose, 6^3-α-D-glucosyl maltotriose and 6^3,6^5-di-α-D-glucosyl maltopentaose were similar and quite rapid, being approximately 20% the rate of hydrolysis of maltose (5). It was thus concluded that 6^3-α-D-glucosyl maltotriose and 6^3,6^5-di-α-D-glucosyl maltopentaose, like isomaltose, are hydrolyzed and digested in both the pig and human small intestine. Consequently, AOAC Methods **2009.01** and **2011.25** should be modified to include an incubation of SDFS with AMG to remove these branched malto-oligosaccharides before HPLC analysis. This step has been seamlessly included into the two AOAC methods, ensuring that the concentration of SDFS is maintained, to assist accurate determination of the HPLC eluates by refractive index measurement.

An essential part of the modification was to demonstrate that the additional incubation with high levels of AMG does not lead to hydrolysis and thus underestimation of other SDFS components, such as FOS, GOS, XOS, RMD, galactosyl-sucrose oligosaccharides or polydextrose.

Experimental

Materials

(**a**) *Chemicals and enzymes.*—D/L-maleic acid (Cat. No. M-0375), bovine serum albumin (Cat. No. A-2153), dimethyl sulfoxide (Cat. No. D-8779), stachyose (Cat. No. S4001-500MG), and sodium azide (Cat. No. S8032) were from Sigma-Aldrich Ltd, Dublin, Ireland. Acetic acid (glacial) GR, sodium hydroxide, and calcium chloride ($CaCl_2 \cdot 2H_2O$) were from Merck, Darmstadt, Germany. Amyloglucoside (E-AMGDF) (EC 3.2.1.3), thermostable α-amylase (E-BLAAM) (EC 3.2.1.1), and protease (E-BSPRT and E-BSPRPD) (EC 3.4.21.14) were from Megazyme International, Bray, Ireland. Industrial methylated spirits (IMS), obtained from Lennox Lab Supply, Naas Road, Dublin, Ireland, is a mixture of ethanol (95%), methanol (5%), and some butanol. In this study, the use of IMS can be replaced by laboratory grade (95–99%) ethanol. The total dietary fiber assay kit (Cat. No. K-TDFR), integrated total dietary fiber assay kit (Cat. No. K-INTDF), total starch assay kit (Cat. No. K-TSTA), and resistant starch assay kit (Cat. No. K-RSTAR) were obtained from Megazyme International Ireland. Amberlite FPA53 (OH–) and Amberlite 200 C (H+) (G-AMBOH or G-AMBH, respectively) were from Megazyme, and in bulk from Rohm and Haas, France S.A.S., Chauny, France.

(**b**) *Polysaccharides.*—Barley β-glucan (medium viscosity; Cat. No. P-BGBM) and wheat arabinoxylan (Cat. No. P-WAXM) were from Megazyme. Partially degraded chicory inulin (Raftilose P-95) was a kind gift from Beneo-Orafti, Tienen, Belgium. Regular maize starch was from Penford Australasia, Lane Cove, NSW, Australia. Hylon VII (ref. 98GH8401), Novelose 330 (ref. AH17529), and Novelose 240 (ref. 96LF10063) were from Ingredion, Bridgewater, NJ. Native potato starch was from Avebe, Foxhol, The Netherlands. ActiStar (enzyme-modified tapioca/cassava starch; U.S. Patent 6,043,229) was from Cargill, Vilvoorde, Belgium. Fiber-Rite and Fibersym were from MGP Ingredients Inc., Atchison, KS. polydextrose (Litesse) was from Danisco, Ardslay, NY, and Fibersol 2 and DE-25 corn syrup (Pinedex) were from Matsutani Chemical Co., Hyogo, Japan. ACS soluble starch (Cat. No. S-9765) was from Sigma-Aldrich. Galacto-oligosaccharides (GO-P90, Lot 0000015274) Purimune High Purity GOS was from Ingredion, Bridgewater, NJ. Xylo-oligosaccharides were from Fugia, Belgium.

(**c**) *Buffer.*—The buffer used was 50 mM sodium maleate (pH 6.0) containing calcium chloride (2 mM) and sodium azide (0.02%).

(**d**) *Dry food and bean samples.*—Dry food samples were obtained and prepared as follows: corn flakes, All Bran, Frosties and Shreddies (Kellogg Co.); Weetabix (Weetabix Co., Kettering, UK); Porridge Oats and oat bran (Flahavan's & Sons Ltd, Kilmacthomas, Ireland); Ryvita dark rye crackers (The Jordans and Ryvita Co., Biggleswade, UK); Carr's table water biscuits (Carr's of Carlisle, Carlisle, UK); and dried organic penne pasta from durum wheat (Meridian Foods, Danbighshire, Wales) were obtained from Tesco Supermarket, Greystones, Ireland. All materials were milled to pass a 0.5 mm screen, using a Fritsch MILL PULVERISETTE 14 (Fritsch, Idar-Oberstein, Germany). Alternative mills that grind materials to pass a 0.5–0.7 mm screen can also be used. Moisture contents were determined for all samples by AOAC Method **925.10**.

(**e**) *Wet food and bean samples.*—Wet bean, vegetable (swede, cauliflower, broccoli, cabbage, carrot, and potato) and fruit samples (ripe banana) were obtained from Tesco Supermarket or a local fruit and vegetable market and prepared as follows: potato was cooked and cooled before lyophilization. Banana was minced with a standard mechanical meat mincer (Porkert, Czech Republic) with a 4.5 mm screen and lyophilized. Fresh swede, cauliflower, broccoli, cabbage, carrot, and potato were peeled (swede and potato), diced to cubes or slices of approximately 1 cm thickness, and lyophilized. Canned red kidney beans (Chivers Ireland Ltd, Coolock, Dublin 15, Ireland), cannellini beans, butter beans, chick peas, garden peas, and sweetcorn were recovered on a food strainer and washed with water. They were then minced (4.5 mm screen) and lyophilized. All bread samples were lyophilized. All lyophilized material was milled to pass a 1.0 mm screen, using a Fritsch MILL PULVERISETTE 14 before analysis.

Equipment and Apparatus

The equipment and apparatus in these experiments were used exactly as described in AOAC Method **2011.25**.

(**a**) *Grinding mill.*—Centrifugal, with 12-tooth rotor and

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0.5 mm sieve, or similar device. Alternatively, a cyclone mill can be used for small test laboratory samples, provided that the mill has sufficient air flow or other cooling to avoid overheating samples.

(**b**) *Digestion bottles*.—250 mL Fisherbrand® soda glass, wide-mouthed bottles with polyvinyl-lined cap (Cat. No. FB73219, equivalent: Cat. No. 02-911-916; Fisher, www.fisher.co.uk/supplies/fisherbrand.html) or 250 mL polypropylene bottles with polypropylene caps.

(**c**) *Fritted crucible*.—Gooch, fritted disk, Pyrex® 50 mL, pore size, coarse, ASTM 40–60 µm (Corning No. 32940-50C® or equivalent; http://www.labplanet.com/corning-crucible-gooch- high-c-50-ml-32940-50c.html). Prepare four for each sample as follows: Ash overnight at 525°C in muffle furnace. Cool furnace to 130°C before removing crucibles to minimize breakage. Vacuum to remove any residual Celite and ash material by using a vacuum. Soak in 2% cleaning solution (Micro-90R; International Products Corp., Trenton, NJ) at room temperature for 1 h. Rinse crucibles with water and deionized water. For final rinse, use 15 mL acetone, and air-dry. Add approximately 1.0 g Celite to dried crucibles and dry at 130°C to constant weight. Cool crucible in desiccators for approximately 1 h, and record mass of crucible containing Celite.

(**d**) *Filtering flask*.—Heavy-walled, 1 L with side arm.

(**e**) *Rubber ring adaptors*.—For use to join crucibles with filtering flasks.

(**f**) *Vacuum source*.—Vacuum pump or aspirator with regulator capable of regulating vacuum.

(**g**) *Water bath(s)*.—Rotary motion, large-capacity (20–24 L) with covers; capable of maintaining temperature of 37 ± 1°C and 60 ± 1°C; equipped with automatic timers for on-off operation or equivalent (e.g., Grant® OLS 200 shaking incubation bath; http://www.bioresearchonline.com/product.mvc/Grant-OLS200-Orbital-Shaking-Water-Bath-0001). Ensure that shaking action/sample agitation in water bath used is sufficient to maintain sample solids in suspension and no residue build-up or rings of sample material form in the digestion bottle during the enzymatic digestions. A linear motion (back and forth shaker) can be used if the bottles are placed at 45° to ensure adequate agitation. If the bottles are vertical or horizontal there will not be sufficient agitation to ensure that the sample remains suspended. Alternatively, a 2 mag Mixdrive 15 submersible magnetic stirrer apparatus (http://www.2mag.de/english/stirrer/multiple/stirrer_multiple_04_mixdrive6_15.html) can be used in a water bath maintained at 37 ± 1°C with a Julabo circulating heater (http://www.julabo.de/). Samples are stirred in digestion bottles, (**b**), with a 7 × 30 mm stir bar at approximately 170 rpm (Figure **2011.25A**). Figures **2011.25A–D** are found in the OMA method.

(**h**) *Balance*.—0.1 mg readability, accuracy, and precision.

(**i**) *Ovens*.—Two, mechanical convection, set at 105 ± 2°C and 130 ± 3°C, respectively.

(**j**) *Timer*.

(**k**) *Desiccator*.—Air-tight, with SiO_2 or equivalent desiccant. Desiccant is dried biweekly overnight in 130°C oven.

(**l**) *pH meter*.

(**m**) *Pipettors and tips*.—Positive displacement 50–200 µL and 5 mL capacity.

(**n**) *Dispensers*.—(*1*) 15 ± 0.5 mL for 78% EtOH or industrial methylated spirits (IMS), 95% ethanol or IMS, and acetone. (*2*) 40 ± 0.5 mL for buffer.

(**o**) *Cylinder*.—Graduated, 500 mL.

(**p**) *Magnetic stirrers and stirring bars*.

(**q**) *Rubber spatulas*.

(**r**) *Muffle furnace*.—525 ± 5°C.

(**s**) *Polypropylene columns*.—(If manual deionization is used.) Sample deionization is achieved using a mixed bed resin of approximately 4 g cation [Ambersep 200C (H+)] and 4 g anion [Amberlite FPA53 (OH–)] in a Bio-Rad Laboratories Econo-Pac™ disposable chromatography column (Cat. No. 732-1010; http://www.bio- rad.com/webroot/web/pdf/psd/literature/Bulletin_9103.pdf) with an Alltech One-Way Stopcock (Cat. No. 211524; Figure **2011.25B**).

(**t**) *Cation and anion exchange guard columns*.—(If online simultaneous deionization and quantitation are used; 7.) Cation and anion exchange guard columns, H+ and CO_2^{3-} forms, respectively (Bio-Rad Laboratories, Berkeley, CA; Cat. No. 125-0118, includes one cation and one anion cartridge), with guard column holder (Bio-Rad Laboratories, Cat. No. 125-0139) to hold the two guard column cartridges in series, cation cartridge preceding anion cartridge.

(**u**) *Liquid chromatograph*.—(*1*) *Pump*.—Capable of precision metering at 0.5 mL/min at up to 10 000 psi. Arrangement for HPLC of manually desalted samples on Waters Sugar-Pak® column (http://www.waters.com/waters/support.htm?locale=?&cid=513789&lid=10069619) is shown in Figure **2011.25C**. If Sugar-Pak column and simultaneous online deionization and quantitation are used, a second pump capable of precision metering at 0.05 mL/min at up to 10 000 psi is needed to add disodium calcium EDTA via zero dead volume t-fitting to the line between the deionization cartridges and the quantitation column (*see* Figure **2011.25D**.

(*2*) *Filled loop injector*.—Equipped with 50 µL injection loop.

(*3*) *LC column*.—Waters Sugar-Pak 6.5 × 300 mm (Part No. WAT085188) or equivalent preceded by Waters Guard Pak LC precolumn inserts (Part No. WAT015209) or equivalent.

(*4*) *Refractive index detector*.—Capable of being maintained at 50°C.

(*5*) *Oven*.—To maintain a column temperature of 90°C.

(**v**) *Data integrator or computer*.—For peak area measurement.

(**w**) *Filters for disposable syringe*.—Polyvinylidene fluoride 0.45 µm, 13 or 25 mm.

(**x**) *Filters for water*.—Polyvinylidene fluoride, 0.45 µm, 47 mm.

(**y**) *Filter apparatus*.—To hold 47 mm, 0.45 µm filter, (**x**); to filter larger volumes of water.

(**z**) *Syringes*.—10 mL, disposable, plastic.

(**aa**) *Syringes*.—Hamilton® 100 µL, 710SNR syringe.

(**bb**) *Rotary evaporator*.—Heidolph Laborota® 4000 or equivalent (http://www.lasallescientific.com/products/heidolph/BAL_LARO4000_Nov06_en1.pdf).

(**cc**) *Thermometer*.—Capable of measuring to 110°C.

(**dd**) *Freeze-drying apparatus*.

Measurement of Dietary Fiber

Four methods were used in this study, AOAC Methods **985.29** (8), **991.43** (9), **2009.01** (1, 2), and **2011.25** (1, 3).

Modification to the Method of Preparation of SDFS

SDFS obtained from high-starch-containing samples such as cereal grains, bread and pasta using the current version of AOAC

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Table 1. HMWDF, SDFS, and TDF values for a range of samples analyzed with AOAC Method 2009.01 with and without additional incubation of SDFS with AMG

Sample	HMWDF (%)	SDFS (%) (–AMG)	SDFS (%) (+AMG)	TDF (–AMG)	TDF (+AMG)
Brennan's Wholegrain brown bread[a]	8.66	1.44	1.08	10.1	9.74
Brennan's Wholemeal brown bread[a]	12.39	1.76	1.66	14.15	14.09
Brennan's White Bread[a]	4.77	2.49	0.45	7.26	5.22
Ryvita Dark Rye Crackers	16.29	6.30	4.70	22.59	20.99
Carr's Water Biscuits	4.20	1.71	1.53	5.91	5.73
Flahavan's Quick oat porridge	9.32	1.15	0.98	10.47	10.3
Oat bran	18.86	0.56	0.54	19.42	19.4
Weetabix	9.82	2.78	2.34	12.60	12.16
Kellogg's Corn Flakes	3.71	2.35	0.08	6.02	3.79
Kellogg's All Bran	26.62	3.86	2.45	30.48	29.07
Frosties Breakfast Cereal	3.10	1.97	0.83	5.07	3.93
Kellogg Shreddies Breakfast Cereal	10.96	3.31	1.97	14.27	12.93
Whole-wheat pasta[a]	9.90	2.77	1.89	12.67	11.79
Carrot[a]	21.81	0.57	0.47	22.38	22.28
Ripe banana[a]	30.18	0.88	0.88	31.06	31.06
Red kidney beans (tinned)[a]	22.80	2.39	2.49	25.19	25.29
Cannellini beans[a]	20.53	2.04	1.70	22.57	22.23
Butter beans[a]	20.16	2.88	2.24	23.04	22.40
Garden peas (petit pois)[a]	29.08	1.37	1.27	30.45	30.35
Chick peas (tinned)[a]	17.98	2.53	1.27	20.51	19.25
Sweetcorn (tinned)[a]	12.69	0.41	0.05	13.1	12.74
Raw swedes[a]	20.27	1.55	1.08	21.82	21.35
Cooked swede[a]	40.41	0.36	0.32	40.77	40.73
Raw cauliflower	21.03	0.48	0.51	21.51	21.13
Cooked cauliflower[a]	23.33	0.41	0.10	23.74	23.43
Broccoli[a]	28.09	0.35	0.40	28.44	28.49
Raw potato[a]	31.74	0.20	0.47	31.94	32.21
Cooked potato[a]	10.66	2.63	0.57	13.29	11.23
Cabbage[a]	28.87	1.30	1.12	30.17	29.99

[a] Samples freeze-dried before analysis. All other samples were analyzed as is. Dietary fiber values are presented as g/100 g on a dry weight basis.

Methods **2009.01** and **2011.25**, contain branched maltodextrins, which are resistant to hydrolysis by the levels of PAA and AMG used in the standard assays. Because these branched maltodextrins are hydrolyzed by the α-glucosidase complex in the small intestine (5), they are not dietary fiber and thus must be removed before determination of SDFS by HPLC. This is achieved as follows: collect half of the aqueous ethanol filtrate from Step **H**(**a**) of Method **2009.01** or Step **I**(**d**) of Method **2011.25** and concentrate to dryness by rotary evaporation. Dissolve the residue in 5 mL of 150 mM HCl to give a pH of 4.5, and transfer to a polypropylene tube. Add 0.1 mL AMG (3300 units/mL), seal the tube, and incubate at 60°C for 1 h and then at about 100°C for 5 min. Remove 2 mL of this solution and deionize through a column of mixed-bed resin [4 g Amberlite® FPA53 (OH^-) resin plus 4 g Ambersep® 200 (H^+)] and proceed according to AOAC Method **2009.01** or **2011.25**.

Results and Discussion

Values of HMWDF, SDFS, and TDF obtained for a range of samples using AOAC Method **2009.01**, with and without additional incubation of the SDFS fraction with AMG at 60°C for 1 h are shown in Table 1. It is evident that in all samples that contain starch, there is a slight reduction in the value of SDFS in samples incubated with AMG. This is due (4, 5) to hydrolysis and removal of small amounts of maltodextrins that resistant hydrolysis in the standard procedure. This is clearly seen in Figure 2, which shows the HPLC patterns of the SDFS fraction of long grain rice with and without additional incubation with AMG. These oligosaccharides, 6^3,6^5-di-α-D-glucosyl maltopentaose and 6^3-α-D-glucosyl maltotriose, resist hydrolysis by PAA and AMG as used in AOAC Method **2009.01**, but they are hydrolyzed by the mucosidal α-glucosidase complex present in the brush boarder lining of pig small intestine. Because a similar enzyme

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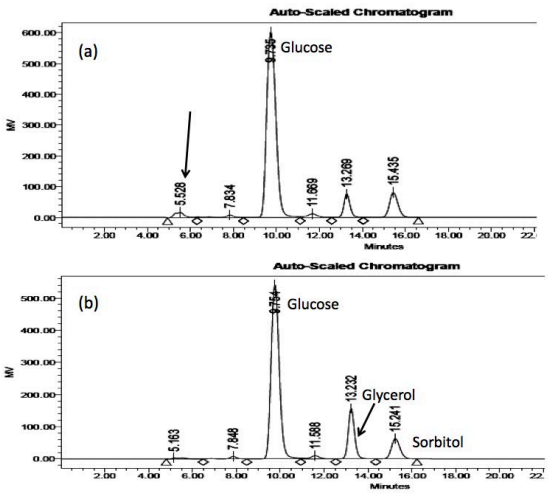

Figure 2. HPLC patterns of SDFS from long grain rice without and with additional incubation with amyloglucosidase. The arrow in (a) shows the presence of $6^3,6^5$-di-α-D-glucosyl maltopentaose. The higher level of glycerol in (b) is derived from the AMG used in the additional incubation.

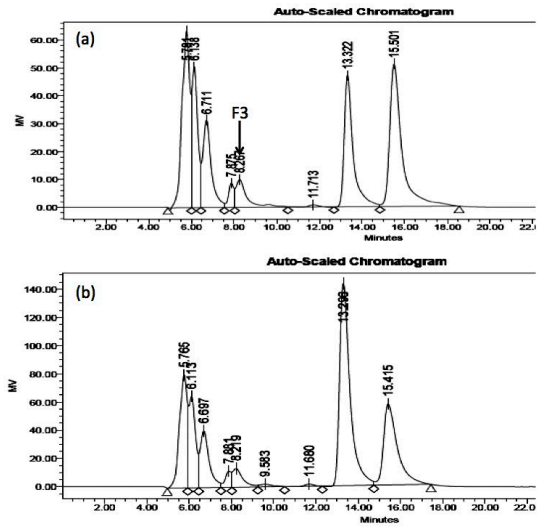

Figure 3. HPLC patterns of Raftilose P-95 recovered as SDFS from dietary fiber analysis according to AOAC Method 2009.01 before (a) and after (b) additional incubation with AMG according to the recommendations. The higher level of glycerol in (b) is derived from the AMG used in the additional incubation. The arrow depicts the point of elution of the trisaccharide β-fructosyl-(2-1)-β-fructosyl-(2-1)-fructose (F3).

complex is known to be present in the brush boarder lining of the human small intestine (6), it can be safely concluded that these oligomers would also be hydrolyzed and absorbed in the human small intestine. Consequently, they should be hydrolytically removed before analysis of SDFS by HPLC.

High levels of AMG are used to hydrolyze the resistant branched maltodextrins; therefore, possible hydrolysis and loss of other nondigestible oligosaccharides commonly found in food materials is possible. To study this, a range of oligosaccharides including FOS, GOS, XOS, RMD, and polydextrose (200 mg) were incubated with AMG under the conditions (60°C, pH 6, 1 h) chosen to hydrolyze the branched maltodextrins, and the samples were deionized, concentrated, and analyzed by HPLC. The results of these studies are shown in Table 2, and HPLC patterns for Raftilose P-95 (FOS) with and without the additional incubation with AMG are shown in Figure 3. Clearly, there is no significant additional hydrolysis of the oligosaccharides, demonstrating that they are stable under the incubation conditions used and that the AMG is free of contaminating enzymes active on these oligosaccharides.

Table 2. Effect of additional incubation with AMG on recovery of various oligosaccharides

	SDFS (% of total sugars in sample)[a]	
Sample details	AOAC **2009.01**	AOAC **2009.01** + AMG
Fibersol 2	83.4	84.9
Polydextrose	85.1	84.8
Raftilose P-95	90.3	89.7
Galacto-oligosaccharides	70.6	71.3
Stachyose (galactosyl-sucrose oligosaccharide)	99.0	99.0
Xylo-oligosaccharides	78.6	77.7

[a] Recovery of oligosaccharides (DP > 3) as a percentage of total sugars separated in the HPLC pattern.

Table 3. Analysis of Kellogg All Bran cereal according to AOAC Method 2009.01; with and without additional incubation of SDFS with AMG

	Sample details		Dietary fiber, % (w/w)		
Analyst	Sample	AMG	HMWDF (IDF + SDFP)	SDFS	TDF
1	All Bran	–	27.51	3.64	31.15
1	All Bran	–	27.10	4.33	31.42
1	All Bran	–	26.68	4.32	31.00
Mean			27.10	4.10	31.19
SD			0.415	0.396	0.212
1	All Bran	+	27.51	3.41	30.92
1	All Bran	+	27.10	3.45	30.54
1	All Bran	+	26.68	3.15	29.82
Mean			27.10	3.34	30.43
SD			0.415	0.163	0.559
2	All Bran	–	26.87	3.92	30.79
2	All Bran	–	26.92	3.81	30.72
2	All Bran	–	26.97	3.86	30.83
Mean			26.92	3.86	30.78
SD			0.050	0.055	0.056
2	All Bran	+	26.87	3.26	30.12
2	All Bran	+	26.92	3.14	30.06
2	All Bran	+	26.97	3.46	30.43
Mean			26.92	3.29	30.20
SD			0.050	0.161	0.198

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The currently proposed modification to AOAC Methods **2009.01** and **2011.25**, through the inclusion of the additional AMG incubation step, affects only the SDFS values because the incubation is performed on this fraction after separation from HMWDF. The repeatability and reproducibility of this modification was studied using a single sample, Kellogg All Bran, which was analyzed in triplicate by two analysts, with and without the additional AMG incubation step; the results are shown in Table 3.

In this study, the SDFS fraction from a given sample was divided into two equal volumes, one half was analyzed by the standard procedure and the second half was concentrated and redissolved in 150 mM HCl, incubated with AMG, then desalted and concentrated for HPLC analysis. The SDFS values (and thus the TDF values) obtained for the three samples by a given analyst and between analysts are very similar. There is a small but significant difference in values obtained for SDFS with and without incubation with AMG. The significance of this difference varies between samples as shown in Table 1 and Figure 2. In Figure 2, the HPLC patterns for SDFS from long grain rice with and without the additional AMG incubation are shown, and the peak shown by the arrow is $6^3,6^5$-di-α-D-glucosyl maltopentaose.

Conclusions and Recommendations

In a previous study (5) it was shown that certain branched maltodextrins that are resistant to hydrolysis in AOAC Method **2009.01** can be hydrolyzed by the mucosidal α-glucosidase complex from the brush boarder lining of the pig (and by inference, human) small intestine. Inclusion of these oligosaccharides in the SDFS fraction leads to an overestimation of this fraction, and thus of TDF, by as much as 2% for some samples rich in starch. These oligosaccharides can be selectively removed by incubation of the SDFS fraction with high levels of AMG. The current study shows that if a sufficiently pure preparation of AMG is used, this incubation leads to no loss of other SDFS components such as FOS and GOS and that the modification to AOAC Method **2009.01** has no deleterious effect on either the repeatability or reproducibility of the method. The author thus recommends that this modification be included in both AOAC Methods **2009.01** and **2011.25**.

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